

CSIE 5452, Spring 2026: Homework 1

B13902022 Yu-Chi,Lai

1 Timing Analysis of the CAN Protocol: Part I (12pt)

1.1

Firstly, set $Q_0 = B_0 = C_1 = 30$, since no other task has higher priority than μ_0 , the terms in the summation is zero.

Hence, the worst-case response time $R_0 = Q_0 + C_0 = 30 + 10 = 40(\text{msec})$.

Iteration	LHS (Q_0)	B_0	RHS	Stop?
1	30	30	30	yes

1.2

Let $Q_1 = B_1 = C_1 = 30$ as the beginnning.

Iteration	LHS (Q_1)	B_1	j	$Q_1 + \tau$	T_j	$\lceil \frac{Q_1 + \tau}{T_j} \rceil$	C_j	RHS	Stop?
1	30	30	0	30.1	50	1	10	40	no
2	40	30	0	40.1	50	1	10	40	yes

Thus, the worst-case response time of μ_1 is $R_1 = Q_1 + C_1 = 40 + 30 = 70(\text{msec})$.

1.3

Let $Q_2 = B_2 = C_2 = 20$ as the beginnning. After 3 iterations, Q_2 converges. Hence, the worst case response

Iteration	LHS (Q_2)	$\lceil \frac{Q_2 + \tau}{T_0} \rceil C_0 + \lceil \frac{Q_2 + \tau}{T_1} \rceil C_1$	RHS	Stop?
1	20	40	60	no
2	60	50	70	no
3	70	50	70	yes

time $R_2 = Q_2 + C_2 = 70 + 20 = 90(\text{msec})$.

2 Timing Analysis of the CAN Protocol: Part II (36pt)

2.1

$R_0 = 1$
 $R_1 = 2$
 $R_2 = 2$
 $R_3 = 3$
 $R_4 = 3$
 $R_5 = 4$
 $R_6 = 5$
 $R_7 = 8$
 $R_8 = 9$
 $R_9 = 9$
 $R_{10} = 10$
 $R_{11} = 19$
 $R_{12} = 19$
 $R_{13} = 20$
 $R_{14} = 29$
 $R_{15} = 29$
 $R_{16} = 30$

2.2

```

1  #include <bits/stdc++.h>
2  using namespace std;
3  signed main() {
4      ifstream fin("input.dat");
5      int n;
6      double tau;
7      fin >> n;
8      fin >> tau;
9      vector<int> priority(n);
10     vector<double> transmission_time(n), period(n);
11     for (int i = 0; i < n; ++i) {
12         fin >> priority[i] >> transmission_time[i] >> period[i];
13     }
14     fin.close();
15     vector<int> res(n);
16     for (int i = 0; i < n; i++) {
17         double block = 0;
18         for (int j = 0; j < n; j++) {
19             if (priority[j] >= priority[i]) {
20                 block = max(block, transmission_time[j]);
21             }
22         }
23         double wait = block;
24         while (1) {
25             double result = block;
26             for (int j = 0; j < n; j++) {
27                 if (priority[j] < priority[i]) {

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28         result += transmission_time[j] * ceil((wait + tau) / period[j]);
29     }
30 }
31 if (result > period[i]) {
32     res[i] = -1;
33     break;
34 }
35 if (result - wait <= 1e-9 && result - wait >= -(1e-9)) {
36     res[i] = transmission_time[i] + wait;
37     break;
38 } else {
39     wait = result;
40     continue;
41 }
42 }
43 }
44 for (int i = 0; i < n; i++) {
45     cout << res[i] << endl;
46 }
47 }
```

- 3 Timing Analysis of Preemptive Fixed-Priority Scheduling (16pt)
- 4 Timing Analysis of TDMA-Based Protocols (12pt)
- 5 MILP Linearization (12pt)
- 6 Signal Packing (12pt)